SARS-CoV-2 Drug Discovery based on Intrinsically Disordered Regions

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Abstract

We aim to discover novel SARS-CoV-2 therapeutics for use in the treatment of COVID-19 patients.

Relevance
- In the US, COVID-19 is responsible for one in nine deaths each day.
- The FDA has approved just one COVID-19 treatment (remdesivir).
- The WHO advises against the use of remdesivir due to lack of definitive evidence.

Specifically, we search for antiviral drugs capable of inhibiting SARS-CoV-2 replication *in vivo*.

Standard Approach
Identify drugs capable of binding to vital *ordered proteins* within the virus. *Ordered proteins* maintain a fixed three-dimensional structure.
Cons: costs, high failure rate, drug resistance, limited targets

Our Approach
Target protein regions which lack a stable structure, known as *intrinsically disordered regions* (IDRs).

We discover **eleven** new SARS-CoV-2 drug candidates.

Introduction

The dynamic shapes of IDRs allow them to bind with a wide range of binding partners. IDRs thereby play significant roles in disease.

SARS-CoV-2 IDRs contribute greatly to the overall function and virulence of the pathogen, and thus are promising targets.

Our Disordered Target
- We focus on the 3C-like protease (3CLpro), whose main role is to cleave polyproteins into their functional parts during replication.
- Specifically, we target the IDR of the N-terminus (residues 1-6), a region involved in the activity of the catalytic site.

We screen a database of drug-like molecules for 3CLpro inhibition using molecular docking and a predictive model.

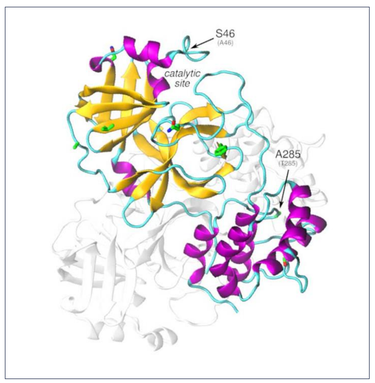


Figure 1. SARS-CoV-2 3CL protease.

Methods

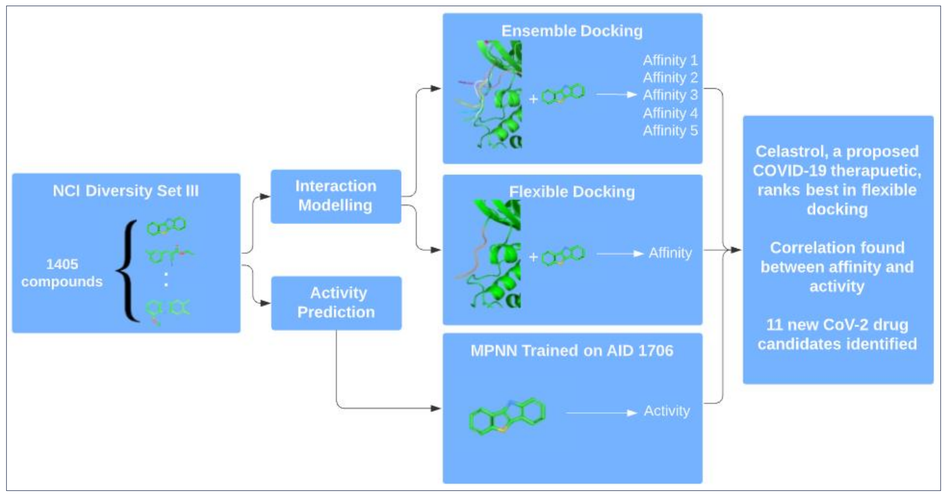


Figure 2. Drug Discovery Approach.

Molecular Docking
Docking employs search algorithms to find the optimal orientation of a ligand onto a protein receptor. Binding affinity measures the strength of this bond and is used here to rank ligands; high affinities suggest that the binding energy will lead to a change in the receptor structure, thereby inhibiting our target.

Standard docking procedures are designed for ordered protein pockets, and thus fail to account for the wide range of IDR flexibility.

We use AutoDock Flexible Receptor (ADFR) to exploit both ligand and receptor flexibility in the search algorithm. By allowing amino acid side chains to rotate, we account for partial receptor flexibility but fail to account for changes in backbone structure. The solution, ensemble docking, involves docking each ligand onto several possible backbone conformations of the IDR. These conformations are generated using loop modelling implemented by MODELLER.

Predictive Model
Expert opinion suggests the cross-verification of docking results using distinct techniques. Thus, we validate and filter our docking results using a predictive model.

The results of bioassay AID 1706, which screens over 290,000 compounds for inhibition of CoV-1 3CLpro-mediated peptide cleavage, are used to train a message passing neural network (MPNN) to distinguish between positive (3CLpro inhibiting) and negative (non-inhibiting) compounds. Due to the high similarity between the two CoV 3CLpros, such a model is likely to make meaningful predictions relevant to CoV-2 3CLpro inhibition.

The model (implemented by Chemprop) is then used to predict activity scores for each previously docked ligand.

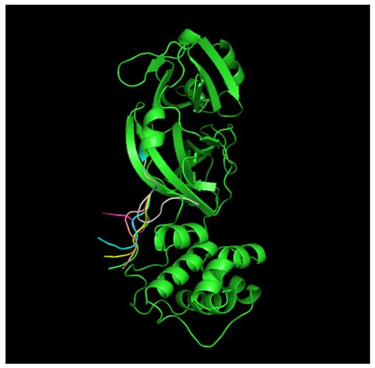


Figure 3. 3CLpro IDR conformations.

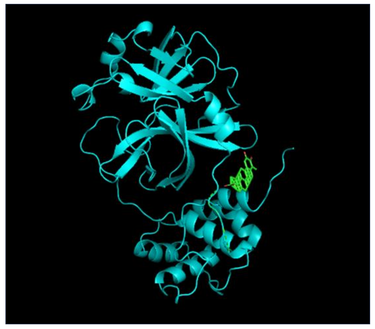


Figure 4. Docked pose of celastrol, the top flexible docking ligand.

Results

Table 1. Top 11 drug candidates in terms of affinity and activity.

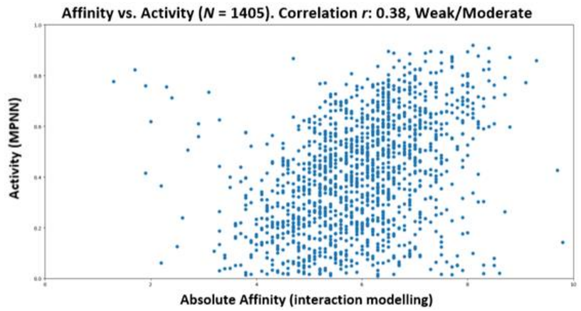
Molecule (NSC)	Activity	Affinity	Active Bioassays
16437	.859	-9.3	Foot-and-mouth disease (FMD) virus
117987	.872	-8.8	
601359	.855	-8.4	Melanoma cell line, Malaria
13294	.825	-8.4	
127133	.908	-8.3	
61610	.823	-8.2	Malaria
107582	.877	-8.1	
128606	.920	-8.0	
211490	.808	-8.0	Hepatitis C virus, Human cytomegalovirus
679525	.894	-8.0	Orthopoxviruses, FMD virus
204232	.800	-7.9	DNA Polymerase Beta

A total of eleven ligands are identified as having both high affinity (absolute affinity ≥ 7.9 kcal/mol) and a high probability of being active (≥ 0.8) according to our MPNN.

These ligands have high probabilities of binding to the IDR, having enough binding energy to deform the 3CLpro, and inhibiting peptide cleavage. Therefore, we deem these ligands promising drug candidates.

Known use cases for these ligands include orthopoxviruses, foot-and-mouth disease virus, human tumors, and malaria. We are currently investigating a potential collaboration to validate the efficacy of the eleven new drug candidates *in vitro*.

Finally, we investigate the link between 3CLpro cleavage inhibition and IDR binding affinity. We find that higher binding affinities to the IDR of the CoV-2 3CLpro weakly-moderately correlate with higher probabilities of cleavage inhibition. This result agrees with the understanding that our IDR helps develop the active site of 3CLpro. Thinking to the future, the result suggests that antiviral drugs targeting the IDR can indeed be therapeutic and that our docking procedure is efficacious in finding such drugs.



Acknowledgements

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References

Please see our paper, "SARS-CoV-2 Drug Discovery based on Intrinsically Disordered Regions," published in the proceedings of the Pacific Symposium on Biocomputing 2021.

Figure 1 retrieved from ORNL's CoV-2 structome site here.